



Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai: 400058, India
(Autonomous College of Affiliated to University of Mumbai)

End Semester Examination

June 2025

Maxi Marks: 100

Class: S.E

Course code: CS206/CE206

Name of the course: Operating Systems

Duration: 3 hours

Semester: IV

Branch: CE/ CSE

Q No		Max Mks	CO
Q1 a	List five services provided by an operating system that are designed to make it more convenient for users to use the computer system. In what cases would it be impossible for user-level programs to provide these services? Explain. OR How does the distinction between kernel mode and user mode function as a rudimentary form of protection (security)? State any two advantages and two disadvantages of both.	10	1-2
Q1 b	Give any three differences between Linker and Loader. Which of the following is True? Justify I. Dynamic linking is done at runtime by an operating system. II. Memory and disk consumption is less in dynamic linking as compared to static linking. III. Linux supports static linking.	10	1-2
Q2 a	What is the cause of thrashing? How does the system detect thrashing? Once it detects thrashing, what can the system do to eliminate this problem? Draw diagram.	10	4-3
Q2 b	Raid configurations of the disks are used to provide A. Fault-tolerance B. High speed C. High data density D. (A) & (B) Consider a 4-drive, 200 GB-per-drive RAID array. What is the available data storage capacity for each RAID level, 0, 1, 2, 3, 4, 5, 6, 10, and 10?	10	5-3

Q3 a	What is the primary purpose of thread synchronization in an OS? How it is implemented? What does the thread scheduler use from the following option/options to decide which thread to run when multiple threads are in the runnable state? Justify your answer i). Thread priority ii). Thread's execution time iii). Thread name iv). Thread creation order Can we call the run() method directly instead of using start ()? Justify your answer	10	2-3		
Q3 b	The following two functions, P1 and P2, share a variable B with an initial value of 2 execute concurrently. <table><tr><td>P1 { C=B-1; B=2xC; }</td><td>P2() { D=2xB; B=D-1; }</td></tr></table> The number of distinct values that B can possibly take after the execution is? Imagine Ram and his friend Sham both have access to the same bank account. They both try to withdraw Rs. 500 at the same time from different ATMs. There is 1500 Rs in the account, and the account balance cannot go below 1000. Whether both withdrawals are possible? Justify your answer.	P1 { C=B-1; B=2xC; }	P2() { D=2xB; B=D-1; }	10	2-3
P1 { C=B-1; B=2xC; }	P2() { D=2xB; B=D-1; }				
Q4 a	Consider a process P running on a CPU. Which one or more of the following events will always trigger a context switch by the OS that results in process P moving to a non-running state (e.g., ready, blocked) and why ? A) P makes a blocking system call to read a block of data from the disk B) P tries to access a page that is in the swap space, triggering a page fault C) An interrupt is raised by the disk to deliver data requested by some other process D) A timer interrupt is raised by the hardware The following C program is executed on a Unix/Linux system: <pre>#include <unistd.h> int main () { int i ; for (i=0; i<10; i++) if (i%2 == 0) fork() ; return 0 ;}</pre> The total number of child processes created is _____.? Explain your answer with neat tree diagram	10	2-3		

Q4 b	Analyze and compare the different free space management techniques used in operating systems—Bit Vector, Linked List, and Grouping. Analyze their trade-offs with respect to memory overhead, speed of allocation, and suitability for large disks.	10	5-3																																													
Q5 a	The Dining Philosophers Problem: Five philosophers are sitting at a round table. In the center of the table is a bowl of rice. Between each pair of philosophers is a single chopstick. A philosopher is in one of three states: thinking, hungry or eating. At various times, thinking philosopher gets hungry. A hungry philosopher attempts to pick up one of the adjacent chopstick, then the other (not both at the same time). If the philosopher is able to obtain the pair of chopsticks (they are not already in use), then the philosopher eats for a period of time. After eating, the philosopher puts the chopsticks down and returns to thinking. Write a monitor code for the dining philosopher problem. OR Create a synchronization strategy using semaphores for the Reader-Writer problem where writers are given priority. Write a code for that and Analyze how your strategy prevents writer starvation.	10	3-3																																													
Q5 b	Consider a system with five processes P0 through P4 and three resources of type A, B, C. Resource type A has 10 instances, B has 5 instances and type C has 7 instances. Suppose at time t0 following snapshot of the system has been taken: <table border="1"><thead><tr><th rowspan="2">Process</th><th colspan="3">Allocation</th><th rowspan="2">Max</th><th rowspan="2">Available</th></tr><tr><th>A</th><th>B</th><th>C</th></tr></thead><tbody><tr><td>P0</td><td>0</td><td>1</td><td>0</td><td>7</td><td>5</td><td>3</td><td rowspan="5">3 3 2</td></tr><tr><td>P1</td><td>2</td><td>0</td><td>0</td><td>3</td><td>2</td><td>2</td></tr><tr><td>P2</td><td>3</td><td>0</td><td>2</td><td>9</td><td>0</td><td>2</td></tr><tr><td>P3</td><td>2</td><td>1</td><td>1</td><td>2</td><td>2</td><td>2</td></tr><tr><td>P4</td><td>0</td><td>0</td><td>2</td><td>4</td><td>3</td><td>3</td></tr></tbody></table> i) What will be the content of the Need matrix? ii) Is the system in a safe state? If Yes, then what is the safe sequence? iii) What will happen if process P1 requests one additional instance of resource type A and two instances of resource type C?	Process	Allocation			Max	Available	A	B	C	P0	0	1	0	7	5	3	3 3 2	P1	2	0	0	3	2	2	P2	3	0	2	9	0	2	P3	2	1	1	2	2	2	P4	0	0	2	4	3	3	10	3-3
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CO Number	CO statement	Marks allotted
CO1	Comprehend the primitive concepts of Operating System services and system Programming functionality.	
CO2	Articulate process scheduling algorithms in the effective execution of processes.	
CO3	Acquaint with efficient process synchronization techniques in the effective execution of programs.	
CO4	Analyze virtual memory management algorithms in effective allocation of main memory usage.	
CO5	Evaluate various algorithms of File Storage & I/O management for performance and quality criteria.	